Cambridge Assessment International Education

Cambridge International AS & A Level

CANDIDATE NAME	
CENTRE NUMBER	
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6007163164

FURTHER MATHEMATICS **9231/33**

Paper 3 Further Mechanics **May/June 2021**

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use $10\,\mathrm{m\,s^{-2}}$.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **16** pages. Any blank pages are indicated.

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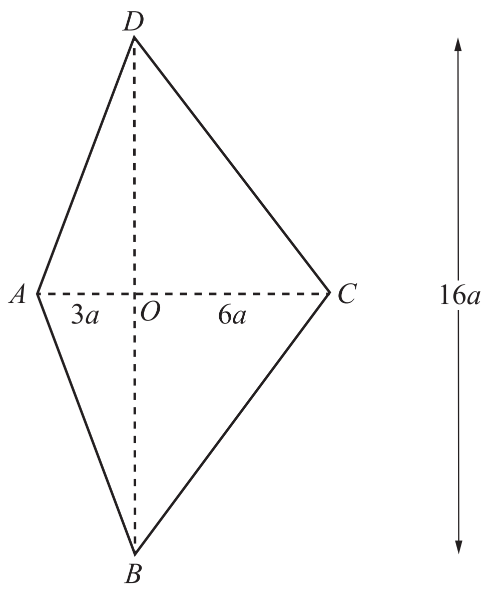
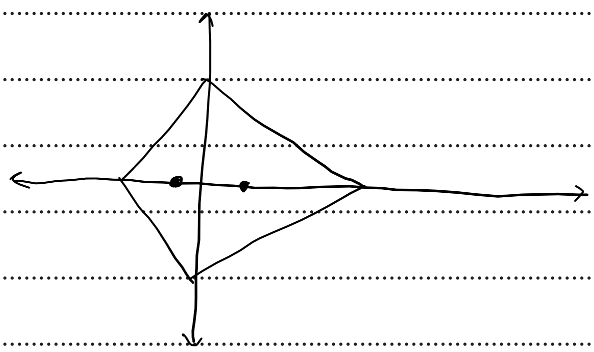


A uniform lamina *ABCD* consists of two isosceles triangles *ABD* and *BCD*. The diagonals of *ABCD* meet at the point *O*. The length of *AO* is $3a$, the length of *OC* is $6a$ and the length of *BD* is $16a$ (see diagram).

Find the distance of the centre of mass of the lamina from *DB*. [3]

As the shape is symmetric about AC

↳ COM lies on AC



$(2a)\cancel{(16a)}(6a)\cancel{(\tfrac{1}{2})}\cancel{(\rho)} + (-a)\cancel{(16a)}(3a)\cancel{(\tfrac{1}{2})}\cancel{(\rho)}$

$= \bar{x}\,\cancel{(16a)}(9a)\cancel{(\tfrac{1}{2})}\cancel{(\rho)}$

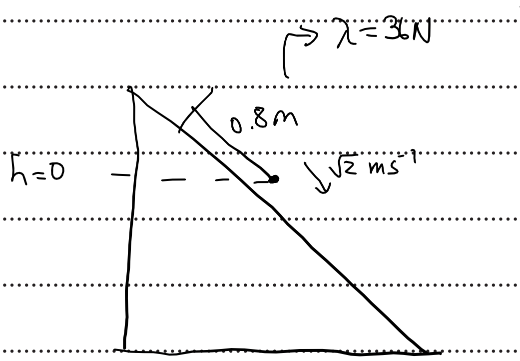
$12a - 3a = 9\bar{x}$

$9\bar{x} = 9a$

$\bar{x} = a \quad \Rightarrow$ COM is $\boxed{a}$ units from DB

2 One end of a light elastic string of natural length 0.8 m and modulus of elasticity 36 N is attached to a fixed point O on a smooth plane. The plane is inclined at an angle α to the horizontal, where $\sin\alpha = \frac{3}{5}$. A particle P of mass 2 kg is attached to the other end of the string. The string lies along a line of greatest slope of the plane with the particle below the level of O. The particle is projected with speed $\sqrt{2}\,\mathrm{m\,s^{-1}}$ directly down the plane from the position where OP is equal to the natural length of the string.

Find the maximum extension of the string during the subsequent motion. [5]



Initially:

$PE = 0$

$KE = \frac{1}{2}(2)(\sqrt{2})^2 = 2J$

$EPE = 0$ as extension $= 0$

$\therefore$ Energy of system $= 2J$

At maximum extension

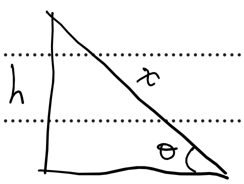
↳ particle is turning around

↳ $\therefore v = 0$

↳ $KE = 0$

$$mgh + \frac{\lambda x^2}{2l} = 2$$

Considering final position



$-h = x\sin\theta$

$$\therefore mgx\sin\theta + \frac{\lambda x^2}{2l} = 2$$

$$-(x)(2)(g)\left(\frac{3}{5}\right) + \frac{36(x)^2}{2(0.8)} = 2$$

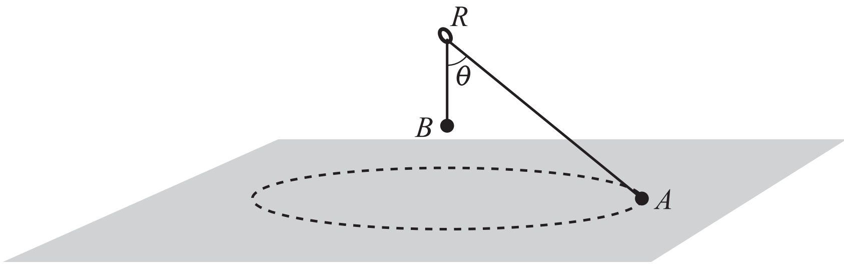
$$22.5x^2 - 12x - 2 = 0$$

$$45x^2 - 24x - 4 = 0$$

$$(15x + 2)(3x - 2) = 0$$

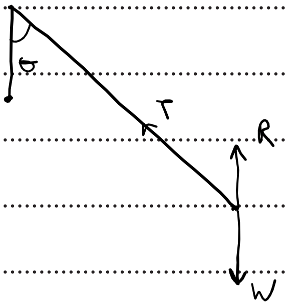
↳ as $x > 0$, $x = \frac{2}{3}m$

3



Particles A and B, of masses $3m$ and m respectively, are connected by a light inextensible string of length a that passes through a fixed smooth ring R. Particle B hangs in equilibrium vertically below the ring. Particle A moves in horizontal circles on a smooth horizontal surface with speed $\frac{2}{5}\sqrt{ga}$. The angle between AR and BR is θ (see diagram). The normal reaction between A and the surface is $\frac{12}{5}mg$.

(a) Find $\cos\theta$. [3]



$T\cos\theta + R = W$

$T\cos\theta = 3mg - \frac{12}{5}mg$

$T\cos\theta = 0.6mg$

$T = B = mg$

$\cos\theta = 0.6$

$\theta = 53.13^\circ$

(b) Find, in terms of a, the distance of B below the ring. [3]

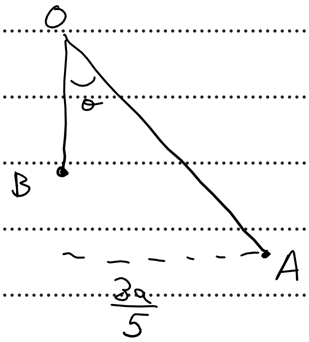
$T\sin\theta = \frac{mv^2}{r}$

$mg\left(\frac{4}{5}\right) = \frac{3m\left(\frac{2}{5}\sqrt{ga}\right)^2}{r}$

$\frac{4g}{5} = \frac{12ga}{25r}$

$\frac{4r}{5} = \frac{12a}{25}$

$r = \frac{3a}{5}$



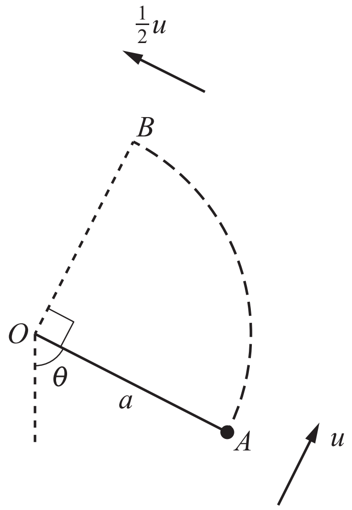
$\sin\theta = 0.8$

$\frac{\frac{3a}{5}}{OA} = 0.8$

$OA = \frac{3a}{4}$

$OB = a - \frac{3a}{4} \boxed{= \frac{a}{4}m}$

4



A particle of mass m is attached to one end of a light inextensible string of length a. The other end of the string is attached to a fixed point O. The particle is initially held with the string taut at the point A, where OA makes an angle θ with the downward vertical through O. The particle is then projected with speed u perpendicular to OA and begins to move upwards in part of a vertical circle. The string goes slack when the particle is at the point B where angle AOB is a right angle. The speed of the particle when it is at B is $\frac{1}{2}u$ (see diagram).

Find the tension in the string at A, giving your answer in terms of m and g. [8]

We can derive 4 equations (for tension and force)
↳ Taking initial speed as $n\ ms^{-1}$:

$T_A = \frac{mn^2}{r} - 2mg + 3mg\cos\theta$ @ A ①

$T_B = \frac{mn^2}{r} - 2mg + 3mg\,(\cos(\theta+90))$ @ B ②

$= \frac{mn^2}{r} - 2mg - 3mg\sin\theta$

$\frac{mu^2}{r} = \frac{mn^2}{r} - 2mg + 2mg\cos\theta$ @ A ③

$\frac{mu^2}{4r} = \frac{mn^2}{r} - 2mg + 2mg\cos(\theta+90)$

$\frac{mu^2}{r} = \frac{4mn^2}{r} - 8mg - 8mg\sin\theta$ @ B ④

Equating ③ & ④, we get:

$$\frac{mn^2}{r} - 2mg + 2mg\cos\theta = \frac{4mn^2}{r} - 8mg - 8mg\sin\theta$$

$$0 = \frac{3mn^2}{r} - 6mg - 8mg\sin\theta - 2mg\cos\theta$$

$$0 = \frac{mn^2}{r} - 2mg - \frac{8}{3}mg\sin\theta - \frac{2}{3}mg\cos\theta \quad (X)$$

→ We know:

$$T_B = 0 = \frac{mn^2}{r} - 2mg - 3mg\sin\theta \quad (Y)$$

By comparing (X) & (Y)

$$3mg\sin\theta = \frac{8}{3}mg\sin\theta + \frac{2}{3}mg\cos\theta$$

$$\frac{1}{3}\sin\theta = \frac{2}{3}\cos\theta$$

$$\tan\theta = 2$$

$$\theta = 63.43^\circ$$

$$T_B = 0 = \frac{mn^2}{r} - 2mg - 3mg\sin\theta$$

$$\frac{n^2}{r} = 2g + 3g\sin\theta$$

$$\frac{n^2}{r} = 2g + \frac{6\sqrt{5}}{5}g$$

$$T_A = \frac{mn^2}{r} - 2mg + 3mg\cos\theta$$

$$T_A = m\left(2g + \frac{6\sqrt{5}}{5}g\right) - 2mg + 3mg\left(\frac{1}{\sqrt{5}}\right)$$

$$= mg\left(2 + \frac{6\sqrt{5}}{5} - 2 + \frac{3\sqrt{5}}{5}\right)$$

$$\boxed{= \frac{9\sqrt{5}}{5}mg}$$

5 A particle P of mass m kg is projected vertically upwards from a point O, with speed $20\,\mathrm{m\,s^{-1}}$, and moves under gravity. There is a resistive force of magnitude $2mv$ N, where $v\,\mathrm{m\,s^{-1}}$ is the speed of P at time t s after projection.

(a) Find an expression for v in terms of t, while P is moving upwards. [6]

Net force $= -mg - 2mv$ (upward)

$a = -g - 2v \quad ms^{-2}$

$\frac{dv}{dt} = -g - 2v$

$\frac{1}{g+2v}\,dv = -\,dt$

$\int \frac{1}{g+2v}\,dv = -\int dt$

$\frac{\ln|g+2v|}{2} = -t + c$

@ $t=0$, $v=20$

$\frac{\ln 50}{2} = c$

$\frac{\ln|g+2v|}{2} - \frac{\ln 50}{2} = -t$

Rearranging for v:

$\ln\left(\frac{g+2v}{50}\right) = -2t$

$e^{-2t} = \frac{g+2v}{50}$

$g + 2v = 50e^{-2t}$

$2v = 50e^{-2t} - g$

$v = 25e^{-2t} - 5$

The displacement of P from O is x m at time t s.

(b) Find an expression for x in terms of t, while P is moving upwards. [2]

$v = 25e^{-2t} - 5$

$\int v\,dt = \int 25e^{-2t} - 5\,dt$

$x = \frac{25e^{-2t}}{-2} - 5t + C$

@ $t = 0,\ x = 0$

$0 = -12.5 + C \Rightarrow C = 12.5$

$\hookrightarrow x = -12.5e^{-2t} - 5t + 12.5$

(c) Find, correct to 3 significant figures, the greatest height above O reached by P. [2]

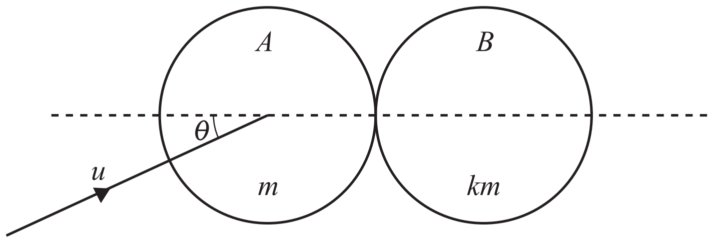
$\hookrightarrow$ greatest height is when $v=0$

$\hookrightarrow e^{-2t} = \frac{1}{5}$

$x = -12.5\left(\frac{1}{5}\right) - 5\left(\frac{\ln 0.2}{-2}\right) + 12.5$

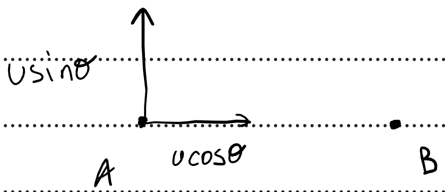
$= 5.98$m

6



Two uniform smooth spheres A and B of equal radii have masses m and km respectively. Sphere A is moving with speed u on a smooth horizontal surface when it collides with sphere B which is at rest. Immediately before the collision, A's direction of motion makes an angle θ with the line of centres (see diagram). The coefficient of restitution between the spheres is $\frac{1}{3}$.

(a) Show that the speed of B after the collision is $\dfrac{4u\cos\theta}{3(1+k)}$. [3]



CoM: $mu\cos\theta = mv_{xA} + kmv_{xB}$

$u\cos\theta = v_{xA} + kv_{xB}$ (1)

Newton's:

$$\frac{v_{xB} - v_{xA}}{u\cos\theta} = \frac{1}{3}$$

$$v_{xB} - v_{xA} = \frac{u\cos\theta}{3} \quad (2)$$

adding (1) & (2)

$$\hookrightarrow (k+1)(v_{xB}) = \frac{u\cos\theta}{3} + u\cos\theta \Rightarrow \boxed{v_{xB} = \frac{4u\cos\theta}{3(1+k)}}$$

70% of the total kinetic energy of the spheres is lost as a result of the collision.

(b) Given that $\tan\theta = \frac{1}{3}$, find the value of k. [6]

$KE_{initial} = \frac{1}{2}mu^2$

$KE_{final} = 0.3\left(\frac{1}{2}mu^2\right) = 0.15mu^2$

$V_B = \frac{4u\cos\theta}{3(1+k)} = \frac{4u}{\sqrt{10}(1+k)}$

$V_{yA} = u\sin\theta = \frac{u}{\sqrt{10}}$

$V_{xA} = \frac{4u\cos\theta}{3(1+k)} - \frac{u\cos\theta}{3} = \frac{u\cos\theta}{3}\left(\frac{4}{k+1} - 1\right)$

$= \frac{u\cos\theta}{3}\left(\frac{3-k}{k+1}\right)$

$= \frac{u(3-k)}{\sqrt{10}(k+1)}$

$V_A = \sqrt{V_{xA}^2 + V_{yA}^2}$

$= \sqrt{\frac{u^2}{10} + \frac{u^2(k-3)^2}{10(k+1)^2}}$

$KE_{final} = \frac{1}{2}(km)\left(\frac{16u^2}{10(k+1)^2}\right) + \frac{1}{2}m\left(\frac{u^2}{10} + \frac{u^2(k-3)^2}{10(k+1)^2}\right)$

$\overset{0.30}{\cancel{0.15}}\,\cancel{m}\cancel{u^2} = \cancel{\tfrac{1}{2}}k\cancel{m}\left(\frac{16\cancel{u^2}}{10(k+1)^2}\right) + \cancel{\tfrac{1}{2}}\cancel{m}\left(\frac{\cancel{u^2}(k+1)^2}{10(k+1)^2} + \frac{\cancel{u^2}(k-3)^2}{10(k+1)^2}\right)$

$0.3 = \frac{16k + (k+1)^2 + (k-3)^2}{10(k+1)^2}$

$3k^2 + 6k + 3 = 16k + k^2 + 2k + 1 + k^2 - 6k + 9$

$k^2 - 6k - 7 = 0$

$(k+1)(k-7) = 0$

$\hookrightarrow$ as $k > 0$, $\boxed{k = 7}$

7 A particle P is projected with speed u at an angle θ above the horizontal from a point O on a horizontal plane and moves freely under gravity. The horizontal and vertical displacements of P from O at a subsequent time t are denoted by x and y respectively.

(a) Use the equation of the trajectory given in the List of formulae (MF19), together with the condition $y = 0$, to establish an expression for the range R in terms of u, θ and g. [2]

$$x\tan\theta - \frac{gx^2}{2u^2}\sec^2\theta = 0$$

$$x\left(\tan\theta - \frac{gx}{2u^2}\sec^2\theta\right) = 0$$

$$\frac{\sin\theta}{\cos\theta} = \frac{gx}{2u^2\cos^2\theta}$$

$$x = \frac{2u^2\sin\theta\cos\theta}{g}$$

$$x = \frac{u^2\sin 2\theta}{g}$$

(b) Deduce an expression for the maximum height H, in terms of u, θ and g. [2]

if range $= \dfrac{u^2\sin 2\theta}{g}$

max H is at $x = \dfrac{\cancel{2}u^2\sin\theta\cos\theta}{\cancel{2}g}$

$$y = \left(\frac{u^2\sin\theta\cos\theta\tan\theta}{g}\right) - \frac{\cancel{g}}{2u^{\cancel{2}}}\left(\frac{1}{\cos^2\theta}\right)\left(\frac{u^{\cancel{4}\,2}\sin^2\theta\cos^2\theta}{g^{\cancel{2}}}\right)$$

$$= \frac{u^2\sin^2\theta}{g} - \frac{u^2\sin^2\theta}{2g}$$

$$= \frac{u^2\sin^2\theta}{2g}$$

It is given that $R = \frac{4H}{\sqrt{3}}$.

(c) Show that $\theta = 60°$. [1]

$$\frac{2\cancel{u}^2 \sin\theta\cos\theta}{g} = \left(\frac{4}{\sqrt{3}}\right)\left(\frac{\cancel{u}^2\sin^2\theta}{2\cancel{g}}\right)$$

$$\cos\theta = \frac{\sin\theta}{\sqrt{3}}$$

$$\tan\theta = \sqrt{3} \Rightarrow \theta = 60°$$

It is given also that $u = \sqrt{40}\,\mathrm{m\,s^{-1}}$.

(d) Find, by differentiating the equation of the trajectory or otherwise, the set of values of x for which the direction of motion makes an angle of less than 45° with the horizontal. [4]

$$\frac{dy}{dx} = \tan\theta - \frac{gx}{u^2}\sec^2\theta$$

$$= \sqrt{3} - \frac{10x}{40}\left(\frac{1}{(1/4)}\right)$$

$$= \sqrt{3} - x$$

$$-1 \leq \sqrt{3} - x \leq 1$$

when

$$-1-\sqrt{3} \leq -x \leq 1-\sqrt{3}$$

$$\boxed{\sqrt{3}-1 \leq x \leq \sqrt{3}+1}$$

Additional Page

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